

Learning That Remembers You

An Open-Source AI-Native Learning Platform and Plugin Architecture
for Longitudinal Learner Modelling at Scale

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Abstract

Traditional learning management systems (LMSs) deliver static, one-size-fits-all content with no longitudinal learner model and no adaptive tutoring. Intelligent tutoring systems that adapt remain either narrow-domain research prototypes or products disconnected from the course-hosting infrastructure most organisations already use. We present **Sudar**, a fully open-source, AI-native learning system released under the Apache 2.0 licence, making three contributions: (1) the **Sudar reference platform (LAMP)**—a working implementation unifying authoring, delivery, and intelligence around a persistent Digital Learner Twin, adaptive sequencing, six multimodal delivery formats, and an AI tutor with longitudinal cross-session memory; (2) the **Adaptive Learning Layer (ALP)**—a novel plugin architecture enabling these capabilities to be deployed as independently installable services on top of existing LMSs; and (3) an **economic analysis** demonstrating the full capability set can be delivered at a per-learner infrastructure cost below \$0.02 per month—exceeding 99% reduction relative to both incumbent commercial LMS licensing fees and proprietary AI provider stacks.

1 — LAMP Reference Platform	2 — ALP Adaptive Learning Layer	3 — Economics Cost Analysis
Open-source implementation: Digital Learner Twin, adaptive sequencing, 6 modalities, longitudinal AI tutor.	Plugin architecture to deploy learner memory, adaptive tutoring, and modality choice on existing LMSs.	Empirically observed 99%+ infrastructure cost reduction vs. incumbent platforms and proprietary AI stacks.

1. Introduction

Learning management systems used by most organisations and universities deliver one-size-fits-all content: the same course for every learner, no memory of prior interactions, and no adaptation of sequence, difficulty, or support based on behaviour or prior knowledge. Research consistently shows that adaptive instruction and intelligent tutoring outperform fixed instruction—a gap quantified by Bloom's two-sigma problem [4], which demonstrated that one-to-one tutoring produces learning gains two standard deviations above classroom instruction. Yet mainstream LMS products do not maintain a longitudinal learner model or provide personalised, memory-aware tutoring [28, 1].

The inference cost collapse of the past four years changes the calculus entirely. AI inference costs have fallen approximately 1,000-fold from 2022 to 2026 [3], making it economically feasible—for the

first time—to deliver a full personalised tutoring session to every learner for less than the cost of a single SMS message. Yet commercial LMS vendors continue to charge \$5–\$20 per user per month, with AI features offered as credits-based add-ons requiring enterprise contracts. This pricing structure excludes community colleges, NGOs, corporate L&D teams, and educational organisations in the Global South.

We address both the architectural and economic gap with **Sudar**, an AI-native learning system that: (1) implements a complete open-source reference platform with learner memory, adaptive paths, and a fully realised multimodal delivery stack; (2) is designed so its core components can be deployed as a layer on top of existing LMSs—the Adaptive Learning Layer (ALP); and (3) demonstrates that this entire capability set can be delivered at infrastructure costs approaching zero using open-weight models and zero-cost TTS.

Contributions:

- **LAMP (reference platform):** A fully open-source, working implementation of an AI-native learning system unifying authoring, delivery, and intelligence around a persistent Digital Learner Twin, adaptive sequencing, six multimodal delivery formats, and an AI tutor with longitudinal cross-session memory.
- **ALP (Adaptive Learning Layer):** A novel plugin architecture for deploying learner memory, adaptive tutoring, and modality choice as independently deployable services on top of existing LMSs—without infrastructure replacement.
- **Economic analysis:** Empirically observed infrastructure cost data demonstrating a 99%+ reduction relative to both incumbent commercial LMSs and proprietary AI provider stacks.

2. Background and Related Work

2.1 Foundational Learner Modelling

Bayesian Knowledge Tracing (BKT), introduced by Corbett and Anderson [8], models a student's latent mastery of individual knowledge components as a hidden Markov process, updating estimates from observed performance. The Digital Learner Twin in Sudar departs from BKT deliberately: rather than modelling discrete knowledge components for narrow curricula, it maintains a profile-level representation of learning style, modality affinity, behavioural engagement signals, and longitudinal tutor context—a practical, LLM-era interpretation of an open learner model [6] with longitudinal persistence, optimised for general-purpose course delivery across domains.

2.2 Adaptive Instruction and ITS

Adaptive learning systems that tailor content and difficulty to the learner improve outcomes compared to one-size-fits-all instruction [26, 2]. Persistent learner models allow systems to adapt over time [22, 6]. Bloom's two-sigma problem [4] provides the foundational motivation: if individual tutoring produces gains two standard deviations above classroom instruction, making high-quality adaptive tutoring universally accessible is one of the highest-leverage interventions available in education.

2.3 Multimodal Learning and Dual Coding

Dual coding theory and multimedia learning principles establish that presenting information across complementary modalities—verbal and visual—supports deeper encoding and accommodates individual differences [17, 7]. Content in Sudar is authored once and delivered in six modalities, with a modality dispatcher that recommends formats per learner profile, operationalising dual-coding principles at scale.

2.4 Conversational and LLM-Based Tutors

Khanmigo (Khan Academy) and Socratic (Google) provide conversational tutoring at scale but operate statelessly within sessions—each interaction begins without knowledge of the learner's history, preferences, or prior struggles. Pardos and Bhandari [22] examined learning gain differences with ChatGPT-based tutoring, noting the stateless nature as a structural limitation. Sudar's AI tutor addresses this gap through longitudinal cross-session memory: every tutor exchange is logged and summarised in the learner's *ai_tutor_context*, so the tutor always knows who the learner is, what they have struggled with, and how they prefer to be taught [22, 26].

2.5 xAPI, LRS, and the Interoperability Gap

The Experience API (xAPI) specification [1] standardises how learning experiences are recorded and reported across systems. xAPI and its Learning Record Store (LRS) ecosystem standardise event reporting but provide no intelligence layer—no learner model, no adaptive sequencing, and no tutoring. ALP provides the missing intelligence layer on top of any xAPI-capable LMS.

2.6 Comparative Analysis

Table 1 compares Sudar with representative systems across key capability dimensions.

Feature	Sudar + ALP	AgentTutor	Khanmigo	Learn Your Way	Typical LMS + AI
Learner model	Longitudinal Twin	Session-scoped	Stateless	Per-session	None / stateless
Tutor memory	Cross-session	Within session	Stateless	Not claimed	Stateless
Open learner model	✓	"Your context"	✗	✗	✗
Modalities	6	Text	Text	Text, slides, audio	Text / video
Augments existing LMS	✓ (ALP)	✗	✗	✗	N/A
Open source	✓ Apache 2.0	✗	✗	✗	Rarely
Cost (1,000/month)	~\$21	N/A	~\$4,000	N/A	\$3,400–\$44,000

Table 1. Comparison of Sudar + ALP with representative systems. Khanmigo individual ~\$4/learner/month; district ~\$833/month for 1,000 learners [37]. See Section 5 for full cost methodology.

3. System Design

3.1 Platform Architecture

The Sudar platform has three surfaces sharing one data layer: **Studio** (admin/creator), **Learn** (learner interface), and **Intelligence** (backend engine). All three share a single source of truth—a Supabase/PostgreSQL instance—for authentication, learner profiles, content, and events. Studio and Learn are Next.js 14 applications (TypeScript strict mode, App Router, Tailwind CSS); Intelligence is a Python 3.11+ FastAPI service. The AI provider layer is abstracted behind an environment-variable-configurable fallback chain (Together AI → OpenRouter → OpenAI → Anthropic).

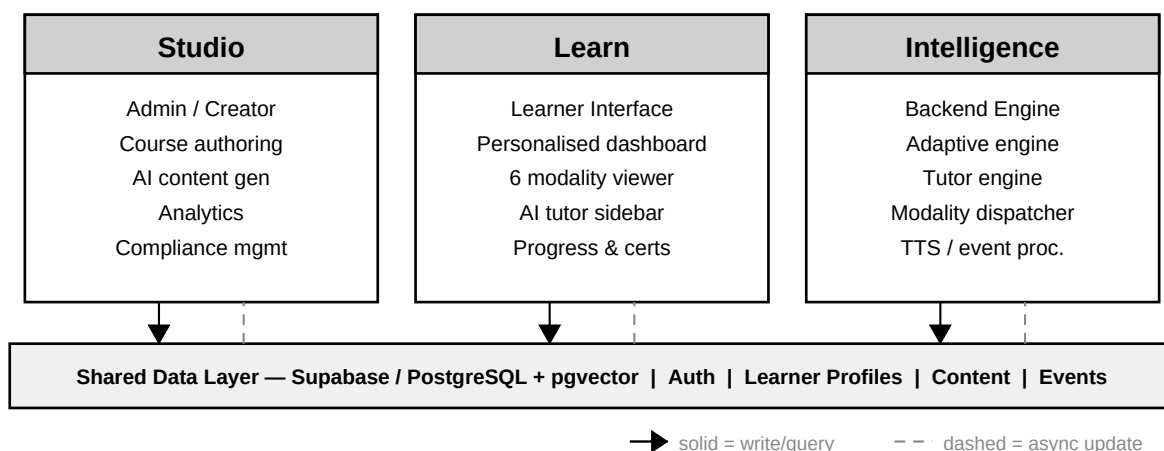
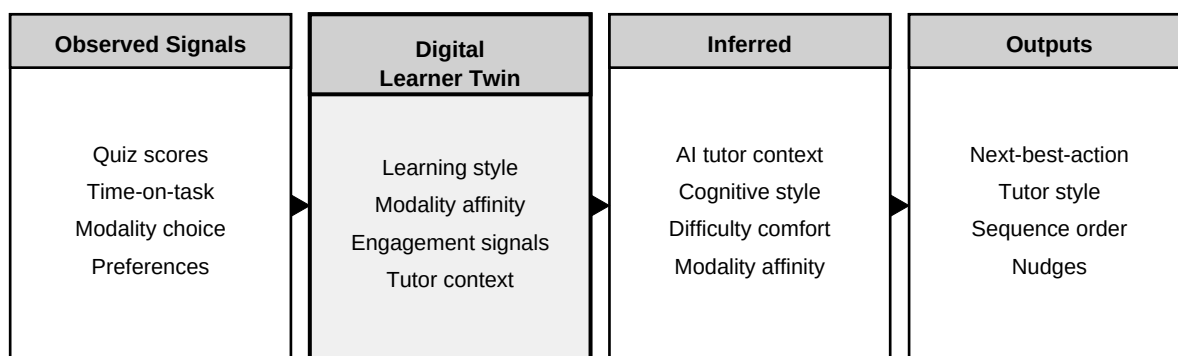


Figure 1. Sudar reference platform — three surfaces over a shared data layer. Solid arrows show write/query flow; dashed arrows show async profile updates.

3.2 Digital Learner Twin

A persistent Digital Learner Twin (DLT) is stored in the *learner_profiles* table. The DLT maintains a unified profile accumulating behavioural, preference, and contextual signals across all sessions and courses. It distinguishes between **observed signals** (quiz scores, time-on-task, modality selections) and **inferred signals** (modality affinity from completion patterns, difficulty comfort from quiz trajectories). A newly created profile is initialised with neutral affinity (0.5 across all formats); an onboarding preference capture seeds it before adaptive behaviour begins. The learner can inspect and correct their own model at any time via the 'Your context' panel—a live, editable view supporting GDPR Articles 15–16 and FERPA compliance.



Cold start: neutral 0.5 affinity → onboarding capture → data-driven personalisation

Figure 2. Digital Learner Twin data model — observed inputs, inferred signals, and adaptive outputs. The twin persists across all sessions and courses.

3.3 AI Tutor with Longitudinal Memory

The AI tutor Sudar is both **reactive** (retrieval-augmented generation over course content to answer learner questions) and **proactive** (generating nudges when struggle or prolonged pause is inferred

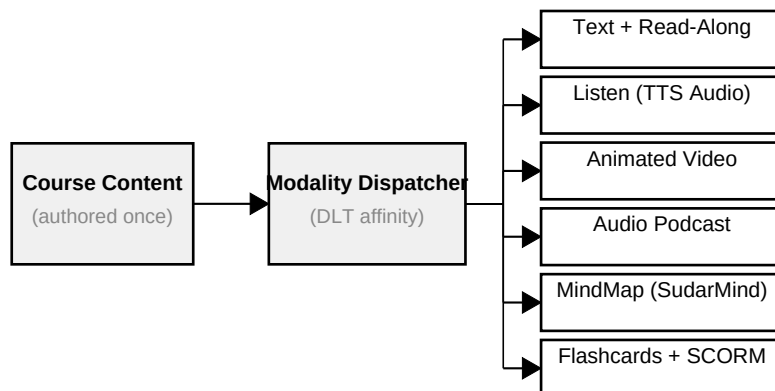
from behavioural signals). Every exchange is logged in *ai_interactions*. The tutor's system prompt includes the learner's memory summary drawn from *ai_tutor_context*, so responses reference prior struggles, match the learner's preferred explanation style, and explicitly connect new material to prior learning. Most AI in LMS implementations are stateless within a session [28]; Sudar maintains memory across sessions and courses.

Key tutor interface features:

- **Text-selection popup** — highlight any passage to invoke 6 pre-built actions (Explain this, Give me an example, Simplify this, Summarise...) or a custom question.
- **Resizable overlay panel and global floating chat** — accessible on any page, not only course pages.
- **Generative block renderer** — tutor can respond with structured artefacts: quiz questions, course recommendation cards, workflow trackers.
- **"My Memory" page** — insights carousel showing what Sudar has learned about the learner, supporting metacognitive reflection.

3.4 Modality-Agnostic Delivery

Content is authored once and delivered in six fully implemented modalities, selectable per learner in real time. A modality dispatcher in the Intelligence layer recommends the next modality from the learner's affinity scores, operationalising dual-coding principles [17, 7].



Evidence: Dual-coding theory [Mayer] · Edge-TTS narration (40+ languages) · Real-time switching

Figure 3. Modality-agnostic delivery stack — content authored once, dispatched across six formats by DLT affinity scores.

The six modalities are:

- **Text with Read-Along** — Sentence-level TTS read-along via Edge-TTS (40+ languages, 300+ voices, zero infrastructure cost). Each sentence is highlighted in place as it plays.
- **Listen (standalone audio TTS)** — Dedicated audio tab for on-demand full-module TTS with optional regeneration.
- **Animated Video** — Scene-by-scene animation rendered entirely in-browser with four styles: kinetic, word-reveal, fade, list.
- **Audio Podcast** — Two-voice dialogue (host + domain expert) with live transcript and jump-to-segment playback.
- **MindMap (SudarMind)** — Interactive concept map via POST `/api/ai/generate-mindmap`. Eight colour families, Bézier curves, module or course scope.

- **Flashcards + SCORM** — On-demand review cards supporting spaced repetition [21, 5]. SCORM packages render in sandboxed iframe; scores write to learning_events.

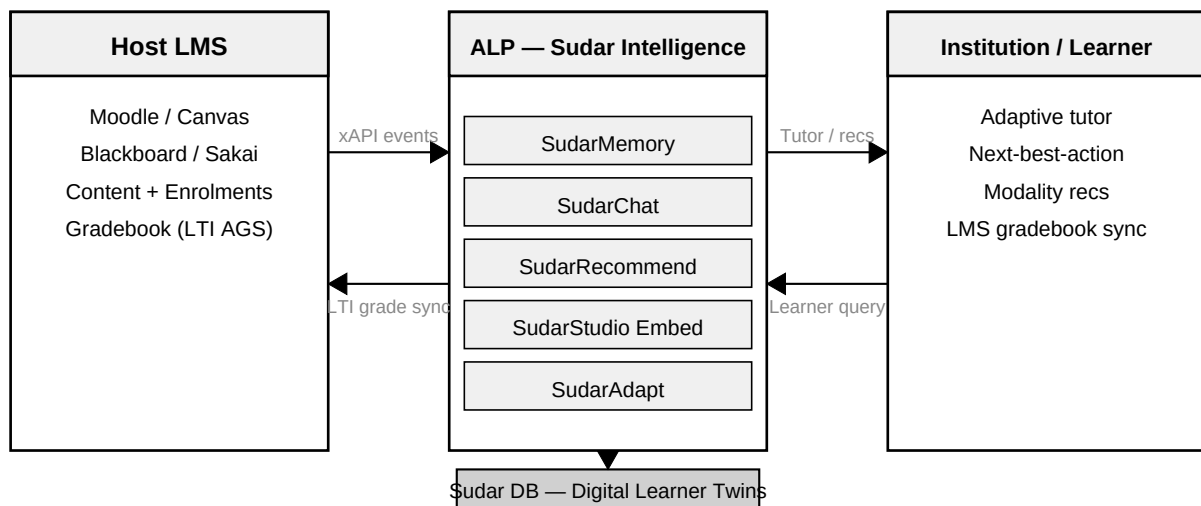
3.5 Adaptive Sequencing and Quiz Architecture

Learning paths support mandatory and optional courses with unlock rules. For optional courses, ordering is adapted by the Intelligence engine. The next-best-action algorithm: (1) filters to enrolled, incomplete items; (2) prioritises mandatory items and approaching due dates; (3) ranks optional items by cosine similarity of skill tags to knowledge gaps multiplied by modality affinity; (4) surfaces the top-ranked item with a human-readable rationale. Five quiz mode archetypes reduce habituation and vary retrieval modality [21]:

- **Standard closed-response** — conventional MCQ or short-answer; appropriate as baseline and for compliance.
- **Predict-then-learn** — learner commits to a prediction before seeing content; activates prior knowledge and increases encoding depth [5].
- **Confidence-tagged** — learner self-reports certainty alongside each answer; supports metacognitive calibration [31].
- **Scenario-fork** — branching case study where learner choices determine the scenario path; models real-world decision-making.
- **Peer-contrast** — presents anonymised responses from other learners before the learner finalises their own.

3.6 Plugin Architecture — The Adaptive Learning Layer (ALP)

The ALP is the paper's most novel architectural contribution: a plugin layer packaging Sudar's capabilities as independently deployable services for existing LMSs. Organisations running Moodle, Canvas, Blackboard, or a proprietary LMS can gain learner memory and adaptive tutoring without abandoning their infrastructure investment. Moodle 4.5 introduced a dedicated AI subsystem [19]; ALP is architecturally aligned with this model. Moodle alone has over 500 million registered users across 233 countries [19, 20].



HTTPS + API keys | LTI 1.3 Advantage | PII-free pseudonymised event stream

Figure 4. ALP integration architecture. The host LMS sends learning events to SudarMemory; ALP returns next-best-action, tutor responses, and modality recommendations via LTI 1.3 Advantage and HTTPS API.

Five independently deployable ALP plugins:

- **SudarMemory** — Foundational layer. Intercepts xAPI or webhook events; maintains a Digital Learner Twin per user. Zero disruption to existing LMS workflows.
- **SudarChat** — Embeds the Sudar AI tutor as a sidebar or modal inside any LMS course page. Every message queries the learner's Twin from SudarMemory.
- **SudarStudio Embed** — Adds 'Generate with AI' to the host LMS course editor for mindmaps, flashcards, audio narration, and animated video storyboards.
- **SudarRecommend** — A next-best-action widget for the LMS dashboard; queries `/api/alp/next-action` with no schema changes to the host database.
- **SudarAdapt** — Integrates with the LMS's conditional activity system to dynamically unlock or reorder optional content.

Security and data isolation: ALP communicates over HTTPS using per-institution API keys. No host LMS student PII is transmitted beyond pseudonymised learning events (user ID hash, event type, timestamp, payload). For Moodle and Canvas, ALP leverages LTI 1.3 Advantage [35] for secure tool embedding, token exchange, and scoped data access.

3.7 Open-Source Extensibility

Sudar's modality delivery stack is designed for community extension and backend substitution. The `VideoScene[]`, `DialogueSegment[]`, and `ModalityVariants` types are provider-agnostic interfaces; any modality backend can be swapped without touching the delivery layer. Video backend options range from the default zero-cost in-browser renderer to Higgsfield [12] (cinema-grade), Wan 2.1 [29] (MIT open-source, 1.3B–14B parameters), and HunyuanVideo [23] (Tencent, Diffusion Transformer). A school in Sub-Saharan Africa can self-host Wan 2.1 on a local GPU at zero per-call cost while the learner experience—Digital Learner Twin, adaptive sequencing, AI tutor—remains identical.

3.8 Intelligence Layer Evolution

The current Intelligence layer is reactive Q&A via RAG, rule-based proactive nudges, and a longitudinal memory updated from quiz outcomes and tutor exchanges. As the `learning_events` table accumulates data, the engine is designed to transition from rule-based heuristics to learned intervention policies. Research demonstrates RL-driven ITS achieves 28.6% better intervention adaptability, 31.2% reduction in recurring errors, and 24.8% lower dropout rates vs. rule-based systems [18].

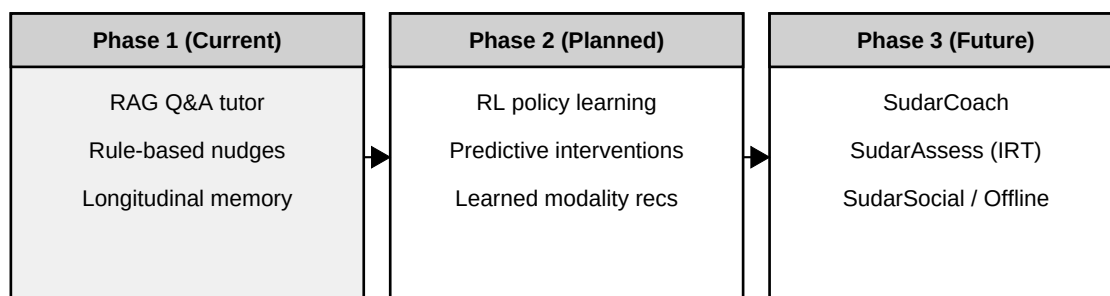


Figure 5. Intelligence layer evolution — from rule-based heuristics (current) to RL-driven policy learning and community intelligence plugins.

3.9 Privacy Architecture and Data Governance

The Digital Learner Twin accumulates behavioural and contextual data across sessions. Sudar's privacy architecture rests on three principles:

- **Data minimisation** — Only behavioural signals directly relevant to learning adaptation are stored: modality affinity, quiz outcomes, time-on-task, and tutor interaction context. Biometric, financial, and demographic data are not collected.
- **User inspection and correction** — The "Your context" panel exposes the full learner model at any time, supporting GDPR Articles 15–16 and FERPA rights.
- **Right to erasure** — The Digital Learner Twin can be reset or deleted at any time (GDPR Article 17 / EU AI Act [10]). Deletion cascades across *learner_profiles*, *ai_interactions*, and *learning_events*.

4. Implementation and Reproducibility

The reference platform is implemented with Next.js 14 (Studio, Learn), Python 3.11+ FastAPI (Intelligence), and Supabase (PostgreSQL + pgvector). AI providers include Together AI (primary), OpenAI, and Anthropic via a configurable fallback chain. TTS uses Edge-TTS (default; zero cost, 40+ languages, 300+ voices) with optional Sarvam AI Indian English neural voices. The system is self-hostable at effectively zero infrastructure cost using free tiers and open-weight model API keys. A live demonstration is accessible at **sudar-learn.vercel.app**. Full schema, API surface, and event model are documented in ECOSYSTEM.md and RESEARCH_FOUNDATION.md in the repository [13].

Table	Key Fields	Purpose
learner_profiles	ai_tutor_context, next_best_action, modality_affinity, streak_days, avg_session_duration, preferences (JSONB)	Digital Learner Twin
learning_events	event_type (module_start/complete, quiz_attempt, video_play, section_heartbeat, ai_tutor_query, modality_switch, drop_off), payload, modality, duration	Behavioural signal log
ai_interactions	user_message, ai_response, context_used, helpful (bool)	Tutor exchange history
enrollments	path/course, progress_pct, due_date, status	Enrolment tracking
certifications	issued_at, expires_at, verification_code	Shareable cert + PDF
skills / learner_skills	skill tags, gap scores	Competency knowledge graph

Table 2. Key schema tables for reproducibility. Full schema and migration files available in *supabase/migrations/* in the repository [13].

5. Economic Accessibility: The Radical Cost Case

The economic dimension of Sudar is not incidental to its design—it is one of its three primary contributions. Operating Sudar at full engagement—10 courses generated, 20 learner review sessions with AI tutor chat, adaptive personalisation, mindmap and flashcard generation, and TTS narration—costs approximately **\$0.20 per week** on the Together AI + Edge-TTS stack. All figures reflect pricing as observed in Q1 2026; the cost floor continues to fall with Llama 4 and equivalent 2026-era models [3].

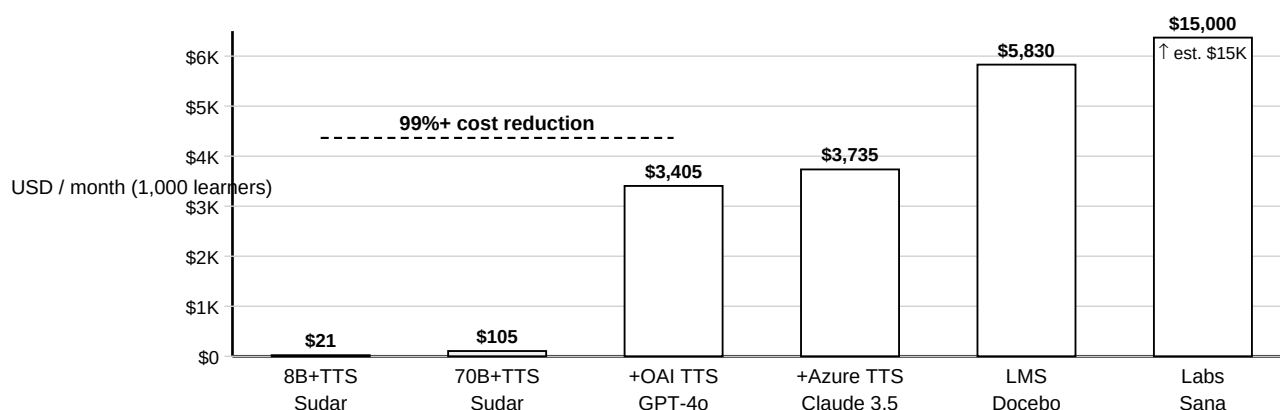


Figure 6. Infrastructure cost comparison per 1,000 learners per month (Q1 2026). Sudar's open-weight stack costs over 99% less than proprietary AI or commercial LMS platforms.

Stack	Per learner / month	Per month (1k)	Annual (1k)
Sudar — Together AI 8B + Edge-TTS	\$0.021	\$21	\$252
Sudar — Together AI 70B + Edge-TTS	\$0.105	\$105	\$1,260
Self-hosted (Ollama + open-weight, GPU)	~\$0*	~\$0*	~\$0*
GPT-4o + OpenAI TTS-1	\$3.41	\$3,405	\$40,860
Claude 3.5 Sonnet + Azure TTS	\$3.74	\$3,735	\$44,820
Docebo (platform licence, low end)	\$5.83+	\$5,830+	\$69,960+
Sana Labs (est., min. 300 seats)	~\$15	~\$15,000	~\$180,000

Table 3. Infrastructure cost comparison (Q1 2026). *Self-hosted cost ~\$0 per API call after hardware provisioning; excludes amortised GPU hardware. Supabase free tier: \$0; Pro: \$25/month. Studio/Learn on Vercel free tier: \$0. Intelligence on Railway/Render: ~\$5–\$10/month at moderate traffic.

5.1 The Inference Cost Collapse

AI inference costs have fallen approximately 1,000-fold from 2022 to 2026 at roughly 10× per year [3], driven by GPU efficiency gains, software optimisation (vLLM, TensorRT-LLM), Mixture-of-Experts architectures, and quantisation. GPT-3-class performance costs \$0.06 per million tokens today versus \$60 in 2021. The arrival of open-weight frontier models (Llama 4 and equivalents in 2026) extends this trajectory further. The cost of delivering a world-class personalised tutor interaction to one learner is now less than \$0.02 per month—below the cost of a single SMS message.

5.2 Education Access and Sudar's Response

As of 2024, 251 million children and youth remain out of school globally [29]. 2.6 billion people remain offline, with mobile internet in Africa costing 14× more than in Europe. Sudar's infrastructure cost of \$0.21/user/month (\$2.52/year) represents a 99.8% reduction for the AI infrastructure layer relative to corporate training benchmarks [\$1,283/employee/year, 32]. For connectivity-constrained scenarios, the relevant deployment path is the offline-first self-hosted variant (Wan 2.1 or a quantised open-weight LLM on a local GPU server)—a planned capability (see Section 7.2) that Sudar's modality backend abstraction and self-hostability are designed to make achievable [29].

6. Evidence and Evaluation Strategy

6.1 Research Foundation

The design is evidence-informed. RESEARCH_FOUNDATION.md in the repository [13] maps each capability to learning-science references: adaptive instruction [26, 2], the two-sigma problem [4], dual coding [17, 7], self-regulated learning [31], formative assessment [5, 21], intelligent tutoring [26, 22], open learner models [6, 15], and learner modelling [22, 6].

6.2 Scope of Current Claims

This is a system and architecture paper. Current claims are: (a) the platform implements all described components at the level of a working, deployed proof of concept, accessible at sudar-learn.vercel.app; (b) each component is grounded in established learning-science evidence; (c) the ALP architecture addresses a gap no widely adopted technology currently fills [28]; and (d) the economic cost structure has been empirically observed and validated against current provider pricing as of Q1 2026.

6.3 Planned Pilot Study

A pilot evaluation is planned following the current build phase, conducted in partnership with one or more institutional partners. Research questions:

- Does Sudar's longitudinal tutor memory improve learner satisfaction and self-reported learning experience relative to a stateless tutor baseline?
- Does adaptive sequencing (next-best-action) reduce time-to-competency compared to fixed-order course delivery?
- Does ALP plugin integration produce measurable engagement improvement (completion rates, session frequency, tutor interaction depth) on an existing Moodle or Canvas installation?

Design: Mixed-methods. Quantitative: pre/post knowledge assessments, module completion rates, time-on-task, quiz score trajectories. Qualitative: semi-structured interviews. **Participants:** 50–200 learners, one partner institution, 4–8 weeks, ≥ 5 courses per learner. IRB/ethics board approval will be obtained prior to data collection. Participants retain the right to withdraw at any time and to request deletion of their Digital Learner Twin.

7. Discussion

7.1 Limitations

- **Absence of empirical efficacy data** — Controlled efficacy data are not yet available; claims about learning gains remain grounded in the underlying learning-science evidence rather than Sudar-specific trials. The pilot study described in Section 6.3 is planned to address this directly.
- **ALP connector implementation status** — The ALP API surface is fully defined and documented in ECOSYSTEM.md; LMS-specific connector implementations for Moodle, Canvas, and Blackboard are in active development.
- **Digital Learner Twin accuracy** — The DLT relies on inferred signals that may not fully capture a learner's actual knowledge state [22, 6]. Modality affinity scores are estimates from behavioural proxies; a learner who selects a suboptimal modality will accumulate affinity scores reflecting choice rather than outcome.
- **Edge-TTS licensing** — Edge-TTS accesses Microsoft's Azure Neural Voice infrastructure without a formal commercial licence; production deployments with paying customers should provision a licensed TTS service (Azure AI Speech, Sarvam AI, or equivalent).

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- **Data portability** — The ability for a learner to export their Digital Learner Twin and import it into another platform is not yet implemented; a formal data export and import API is in the roadmap.
 - **Learner twin interpretability** — Educators and administrators may not understand or trust AI-generated inferences about students. Roumpas et al. [36] provide an ethical framework for cognitive-aware human-AI interaction in multimodal adaptive systems; full alignment—including formal auditability of AI-generated inferences—remains future work.

7.2 Future Work

Planned work includes: LMS-specific connector implementations for ALP (Moodle 4.5+, Canvas, Blackboard); completing SudarPlay (gamified modality) and SudarFeed (social feed); developing the predictive intervention layer with RL-based policy learning; running controlled efficacy studies; releasing SudarCoach, SudarAssess, and SudarCompliance as first-party Intelligence plugins; building the community plugin ecosystem; supporting enterprise features (SSO, HRIS integration); and developing an offline-first variant for connectivity-constrained environments. Digital Learner Twin data portability (cross-platform export/import API) is a planned capability.

SudarSocial is in early development as an integration with the open-source virtual world platform WorkAdventure, enabling peer learning orchestration within persistent collaborative learning environments. The Digital Learner Twin will inform peer-matching algorithms—supported by socioconstructivist learning theory [30].

8. Conclusion

We presented Sudar, an AI-native learning platform making three contributions: (1) a reference platform (LAMP) combining authoring, delivery, and intelligence around a Digital Learner Twin, adaptive sequencing, a complete multimodal delivery stack, and an AI tutor with longitudinal cross-session memory—open source, deployed, and accessible at <https://sudar-learn.vercel.app>; (2) the Adaptive Learning Layer (ALP), a plugin architecture with the potential to reach the hundreds of millions of Moodle, Canvas, Blackboard, and Sakai users [19] without requiring LMS replacement; and (3) an empirically observed economic model reducing the infrastructure cost of AI-native personalised learning to less than \$0.02 per learner per month—viable at the scale of national education systems.

The inference cost collapse [3] is the macro condition that makes all of this possible. Bloom's two-sigma problem [4] has been known since 1984; the economic barrier to solving it at population scale has only recently expired. The paper is honest about what Sudar does not yet have: controlled efficacy data, production LMS-specific connectors, and a completed offline-first deployment path. These are the next milestones, not reasons to delay the architectural and economic contributions presented here.

Call for the Community — The reference implementation is open source (Apache 2.0), grounded in a broad learning-science evidence base, and designed as a bedrock upon which the community can build modalities, intelligence plugins, and LMS connectors. Institutions, researchers, LMS administrators, and open-source contributors are invited to engage: use Sudar in your courses, build ALP connectors for your LMS, extend the modality stack, and contribute to the pilot study programme. The goal is a world where adaptive, memory-aware learning is not a privilege of well-funded institutions but open infrastructure—like the web itself.

Acknowledgments

Sudar is the author's independent project, designed and implemented as a solo effort across the Studio, Learn, and Intelligence surfaces. The system and research are available at <https://github.com/Dhanikesh-Karunanithi/Sudar>. Institutions, organisations, and open-source contributors are invited to collaborate on pilots, plugin integrations, and community

extensions. Contact: connect@dhanikeshkarunanithi.com

Appendix A: Schema Summary

Key tables for reproducibility. Full schema, migration files, and event model: ECOSYSTEM.md and supabase/migrations/ in the repository [13].

Table	Key Fields	Purpose
learner_profiles	ai_tutor_context (JSON), next_best_action (JSON), modality_affinity (JSON), course_id, streak_days, avg_session_duration	Profile and context
learning_events	event_type (module_start, module_complete, quiz_attempt, video_play, pause, replay, session_heartbeat, ai_tutor_opportunity)	Behavioural logging
ai_interactions	user_message, ai_response, context_used, helpful (bool)	Tutor history
enrollments	path/course, progress_pct, due_date, status	Enrolment tracking
certifications	issued_at, expires_at, verification_code	PDF cert links
skills / learner_skills / skill_tags	skill_tags, gap scores	Competency knowledge graph
learner_performance / institution	institution-aware KPI and grade tracking	Analytics / compliance

Appendix B: Infrastructure Cost Reference

All figures observed Q1 2026. AI provider pricing changes rapidly; verify current rates at time of deployment.

Stack	Per learner / month	Annual (1k learners)
Together AI 8B + Edge-TTS	\$0.021	\$252
Together AI 70B + Edge-TTS	\$0.105	\$1,260
Self-hosted (Ollama + open-weight, institutional GPU)	~\$0 per call*	~\$0*
GPT-4o + OpenAI TTS-1	\$3.41	\$40,860
Claude 3.5 Sonnet + Azure TTS	\$3.74	\$44,820
Docebo (platform licence, low end)	\$5.83	\$69,960+
Sana Labs (est., min. 300 seats)	~\$15	~\$180,000

*Self-hosted ~\$0 per API call after hardware provisioning; excludes amortised GPU hardware. AI infrastructure only; hosting excluded. Supabase free tier: \$0; Pro: \$25/month.

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